

Sparsh Gupta

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EDUCATION

Bachelor of Technology in Computer Science and Engineering

2021–2025

Galgotias University, Greater Noida, India

- Cumulative Grade Point Average (CGPA): **8.57/10.0**

Senior Secondary Education (Science Stream)

2020–2021

Ryan International School, Shahjahanpur, India

- Percentage Score: **81%**

TECHNICAL SKILLS

- **Programming Languages:** Python, Java, SQL, HTML/CSS, C, C++
- **Data Science & Machine Learning:** Natural Language Processing (NLP), Deep Learning, Computer Vision, Generative AI, Model Fine-Tuning, Feature Engineering, Data Processing Pipeline, Statistical Analysis, Hypothesis Testing, Agentic AI Systems, Large Language Models (LLM)
- **Data Analysis & Visualization:** NumPy, Pandas, Matplotlib, Seaborn, Plotly, Cufflinks, Manim, Microsoft Excel, Jupyter Notebook, Power BI
- **AI Frameworks:** LangChain, LangGraph, LLM Integration, Text-to-Speech Models, Retrieval Augmented Generation (RAG), Autonomous AI Agents
- **ML Libraries & Tools:** Scikit-learn, TensorFlow, PyTorch, OpenCV, Flask, FastAPI, HuggingFace Transformers, Git, GitHub, Docker, Linux, AWS (S3, EC2), Google Cloud Platform (GCP)

WORK EXPERIENCE

Data Scientist

July 2024 - Present

Whilter.AI, Gurugram, India

- Engineered scalable data pipelines using Python and cloud services, collaborating with cross-functional teams to perform comprehensive exploratory data analysis for model optimization that improved processing efficiency by 30%.
- Finetuned and implemented advanced machine learning models for **text-to-speech conversion**, enhancing synthesized speech quality by 40% through specialized dataset retraining and parameter optimization.
- Architected cloud-native solutions integrating AWS S3 storage and Google Cloud Platform services to streamline model training workflows, reducing deployment time by 25%.
- Pioneered a production-ready conversational Voice AI ecosystem for global airline operations, automating high-volume flight inquiries and reducing manual call-handling volume.

PROJECTS

AI-Powered Virtual Assistant for Airlines

- Designed and deployed an AI-powered voice bot for airline operations, building end-to-end pipelines for model hosting, agent development, and productionization. Implemented tool-call integrations for airline-specific queries and low-latency real-time transcription. Fine-tuned TTS and LLM models and adapted Whisper-based STT for language-specific accuracy. Optimized system latency to 600 ms, improving responsiveness, fault tolerance, observability, monitoring, and CI/CD-driven automated deployments across multilingual production environments and reduced call-handling errors while improving user satisfaction.

Professional Network Profile Analysis System

- Developed a Python-based data extraction and analysis tool for LinkedIn and Twitter profiles with 98% accuracy rate. Leveraged LangChain framework for advanced semantic information processing and integrated Pinecone vector database for high-performance similarity-based profile search and retrieval, enabling sub-second query response times.

Advanced Text-to-Speech Model Fine-Tuning

- Developed a state-of-the-art transformer-based TTS model trained on 100+ hours of curated speech data to achieve zero-shot voice synthesis capabilities. Implemented an end-to-end training and optimization pipeline encompassing audio preprocessing, model architecture customization, and inference optimization with very low latency with in 100 ms, resulting in significantly enhanced voice naturalness and multilingual capabilities.

AI-Driven Hyper-Personalized Engine

- Architected an AI-powered dynamic template and message generation system for hyper-personalized insurance renewal communications, improving customer response rates by 28%. Implemented automated image processing algorithms to overlay customer-specific data on branded creative templates, enhancing brand recall metrics and driving measurable increases in customer engagement.

CERTIFICATIONS

- **Python for Data Science and Machine Learning** (Udemy) - Advanced data manipulation, analysis, and modeling techniques
- **LangChain: Develop LLM Powered Applications** (Udemy) - Large language model implementation and integration
- **LangGraph: Develop LLM Powered AI Agents** (Udemy) - Autonomous AI agent development
- **Machine Learning A-Z: AI, Python & R + ChatGPT Prize** (Udemy) - Deep Learning Concepts